

Digital Material Passport

ID **DMP-METAL-002** - Version **1.0.0**
Issue Date **2025-05-14** - Certificate Type **EN 10204 3.1**

Manufacturer		Customer	Business Transaction	
ACME Metal Works GmbH Industrial Park 123 52066 Aachen DE quality@acme-metal.example.com		Engineering Solutions Ltd. Tech Park Way 45 Cardiff CF14 5DU GB procurement@engisolutions.example.com	Order	PO-78902 · 2000 kg Pos. 10 · 2025-04-21
			Delivery	DN-56790 · 2000 kg Pos. 1 · 2025-05-13 Internal Order ID: SO-12346
Subcustomer American Heavy Industries Inc. 5000 Industrial Blvd Building C Detroit, MI 48201 US materials@heavyind.example.com		Goods Receiver Canadian Construction Corp 777 Construction Ave Toronto, ON M5H 2N2 CA logistics@canconstruct.example.ca		

Product

Product Name	Structural Steel S420N	Name (EN)	1.8902
Batch ID	H-10988-01	Shape	Plate - Width 2000 mm - Thickness 25 mm - Length 6000 mm
Surface Condition	Hot-rolled	Product Norm	EN 10025-3 (2019)
Weight	2000 kg		
Production Date	2025-05-10		
Country of Origin	DE		

Heat Treatment

Process: **Normalizing** - Lot: NORM-2024-0823-A12 - Furnace: NF-LINE-02 - Date: 2024-08-23

Stage	Temperature	Duration	Cooling	Atmosphere
Austenitizing	920 C	60 min		Air
Cooling	20 C		Air	

Chemical Analysis

Heat Number **H-10988** - Melting Process **EAF+LF+VD** - Casting Method **ContinuousCasting** - Casting Date **2025-05-09** - Sample Location **Ladle**

Symbol	C	Mn	Si	P	S	CEV
Unit	%	%	%	%	%	%
Min						
Max	0.20	1.60	0.50	0.025	0.015	0.44
Actual	0.16	1.48	0.28	0.016	0.010	0.41

CEV = C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15: 0.41%

Mechanical Properties

Property	Symbol	Actual	Minimum	Maximum	Method	Status
Tensile Strength Test temperature: 20°C Specimen: 1/3R, L						EN ISO 6892-1 ✓
No.	1	2	3	Average	Min	Max
Value [MPa]	558	562	560	560	558	562
						Std Dev
						2
						Spec Min
						520
						Spec Max
						680
Yield Strength Test temperature: 20°C, 3 specimens tested						EN ISO 6892-1 ✓
No.	1	2	3	Average	Min	Max
Value [MPa]	442	445	448	445	442	448
						Std Dev
						3
						Spec Min
						420
Elongation after fracture Gauge length 5.65#S ₀ , 3 specimens tested						EN ISO 6892-1 ✓
No.	1	2	3	Average	Min	Max
Value [%]	23	24	25	24	23	25
						Std Dev
						1
						Spec Min
						19
Reduction of Area¹						EN ISO 6892-1 ✓
No.	1	2	3	Average	Min	Max
Value [%]	60	62	64	62	60	64
						Std Dev
						2
						Spec Min
						50

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Charpy V-notch Impact Energy ²								EN ISO 148-1	✓
Specimen: Top, T-L									
No.	1	2	3	Average	Min	Max	Std Dev	Spec Min	
Value [J]	56	58	60	58	56	60	2	40	

Charpy V-notch Impact Energy ²								EN ISO 148-1	✓
Specimen: Bottom, T-L									
No.	1	2	3	Average	Min	Max	Std Dev	Spec Min	
Value [J]	54	57	59	56.7	54	59	2.5	40	

Brinell Hardness									EN ISO 6506-1			✓
Ball: 10mm, Force: 3000kg, 5 indentations												
No.	1	2	3	4	5	Average	Min	Max	Std Dev	Spec Min	Spec Max	
Value [HBW]	183	185	187	184	186	185	183	187	1.58	150	220	

Vickers Hardness									EN ISO 6507-1		✓
10 kg load, 5 indentations											
No.	1	2	3	4	5	Average	Min	Max	Std Dev	Spec Min	Spec Max
Value [HV10]	192	195	198	194	196	195	192	198	2.35	160	230

Rockwell Hardness	HR	18 HRC	22 HRC	EN ISO 6508-1	✓
Elastic Modulus	E	210 GPa		EN ISO 6892-1	✓
Strain Hardening Exponent	n	0.18		ASTM E646	✓
Plastic Strain Ratio	r	1.2	1.0	EN ISO 10113	✓
0.2% Proof Strength ¹				EN ISO 6892-1	✓

No.	1	2	3	Average	Min	Max	Std Dev	Spec Min	
Value [MPa]	428	430	432	430	428	432	2	400	

¹ 3 specimens tested ² Test temperature: -20°C

Validation

We hereby certify that the material described above has been manufactured and tested in accordance with the requirements of EN 10204:2004 type 3.1 and the specified standards.
The results comply with the requirements.

Validated by **Johann Weber**, Quality Inspector, Quality Assurance - 2025-05-14